Illustrative Texas-Style Outage Scenarios for RAG Testing

SYNTHETIC EDUCATIONAL DOCUMENT — NOT AN OFFICIAL SOURCE

Purpose: Provide background material for retrieval and chatbot testing in the SmartGrid AI prototype.

This document contains fictional scenarios created solely to test question answering and retrieval in a SmartGrid AI prototype. The scenarios are not historical Texas events, do not describe actual customers or utilities, and must never be cited as real outage records.

Scenario A: Summer heat and high demand
A fictional region experiences several days of high temperatures. Cooling demand rises in the late afternoon. A demand forecast indicates a higher-than-usual peak. The system operator reviews available generation, planned maintenance, weather forecasts, and reserve levels. The scenario illustrates why high demand should be assessed together with available supply and transmission conditions.

Scenario B: Thunderstorm and local feeder damage
A fictional thunderstorm causes a tree branch to contact a local distribution line. Protective equipment isolates the fault and a group of customers loses service. Bulk generation remains available. A utility crew inspects the area, removes the hazard, repairs equipment, and restores service after safety checks. The scenario illustrates the difference between a local distribution outage and a bulk-grid shortage.

Scenario C: Winter cold and equipment problems
A fictional cold spell increases heating demand. At the same time, one generating unit becomes unavailable due to an equipment problem. Operators reassess the supply-demand balance and available reserves. The scenario illustrates correlated weather and equipment risks without implying that any particular emergency action is inevitable.

Scenario D: Flooded substation access
A fictional flood makes a road to a substation inaccessible. The utility restricts access until conditions are safe and coordinates inspection and repair. Restoration takes longer because crews cannot immediately reach the site. The scenario illustrates how access and safety can affect outage duration.

Scenario E: Customer preparedness
A fictional household receives notice of possible severe weather. The household charges phones, prepares flashlights and water, checks on a neighbor who uses powered medical equipment, and reviews the utility's official outage-reporting process. The scenario illustrates practical preparedness without replacing local instructions.

Questions for testing
- Can high demand occur without an outage?
- How can a local feeder fault differ from a generation shortage?
- Why might flooding delay restoration?
- Why should a predictive risk score not be described as an official alert?
- Which sources should customers consult for current outage information?

End of synthetic document. Verify operational facts with current official sources before real-world use.